



Original article

A continuous authentication scheme for zero-trust architecture in industrial internet of things

Tao Wan^a, Buhai Shi^{a,ip}, Huan Wang^{b,*}^a School of Automation Science and Engineering, South China University of Technology, Guangzhou 510641, China^b Industrial Technology Research Center, Guangdong Institute of Scientific & Technical Information, Guangzhou 510000, China

ARTICLE INFO

Keywords:

Industrial internet of things authentication

Elliptic curve Cryptography

Physical unclonable functions

ABSTRACT

Industrial Internet of Things (IIoT) allows users to access industrial devices and their data, but it also poses certain challenges to industrial data security. Authentication protocols are highly effective security techniques for protecting industrial data. This paper establishes a zero-trust architecture in the IIoT and proposes an authentication protocol suitable for zero-trust IIoT. The proposed scheme utilizes physical unclonable functions (PUF) for device authentication. Initial device authentication employs PUF to verify identity and establish session keys before session initiation, while continuous authentication verifies device location during the session to ensure that authenticated devices remain unaltered. Meanwhile, the scheme integrates three-factor authentication for user verification, ensuring secure user access. The proposed scheme establishes secure session key for users, gateways and IIoT devices, effectively guaranteeing the security of subsequent communications. Formal security analysis proves the security. Additionally, detailed informal security discussions demonstrate that the scheme can withstand known attacks and meet design objectives. Furthermore, performance evaluation reveals that the proposed scheme incurs low costs while providing enhanced security.

1. Introduction

The Internet of Things (IoT) interconnects various smart devices via the Internet, enabling the flow and sharing of information through intelligent sensing technologies. IIoT systems, as a significant application of IoT technology, are predominantly utilized in industrial domains like energy, manufacturing and power [1–4]. Although IIoT technology accelerates the intelligent upgrading of traditional industries and fosters economic growth, IIoT systems confront significant information security issues, particularly concerning communication confidentiality and data integrity. Industrial data typically comprises sensitive information, including personal identity data, control commands and industrial production metrics, where any breach could result in substantial economic losses for enterprises. The open nature of networks makes IIoT systems particularly susceptible to breaches of communication confidentiality and data integrity via methods like eavesdropping, tampering, or spoofing attacks. Consequently, the foundational prerequisite for IIoT security is the protection of communication confidentiality and data integrity within these systems. Authentication and key exchange protocols are vital mechanisms for ensuring secure communication over

public channels, facilitating mutual identity verification and key negotiation while adhering to the "one-time pad" encryption standard.

Recently, the zero-trust security model has shown tremendous potential in safeguarding complex IoT systems [5–7]. The zero-trust concept provides a novel approach to reimagining network security architecture. It challenges the traditional perimeter-based network security design, rejecting the assumption that entities within the perimeter are inherently secure and follows the principle of "never trust, always verify," by continuously validating identities, assessing device legitimacy and determining access permissions. The core of zero trust is the transition from network-centric security architectures to identity-centric security frameworks. This paper introduces a zero-trust architecture for the IIoT, where authentication secures users, devices, or services connecting to the network before industrial entities access resources. Verified entities are granted customized resources, mitigating potential attacks from malicious entities aiming to disrupt production processes. The control plane constitutes the core of the zero-trust architecture, enabling identity authentication and access control. Within the control plane, users, devices and applications are authenticated, and access permissions are adjusted based on risk assessment outcomes. The data

* Corresponding author.

E-mail address: wanghuan6@email.szu.edu.cn (H. Wang).<https://doi.org/10.1016/j.aej.2025.03.012>

Received 24 January 2025; Received in revised form 26 February 2025; Accepted 4 March 2025

Available online 20 March 2025

1110-0168/© 2025 The Authors. Published by Elsevier B.V. on behalf of Faculty of Engineering, Alexandria University. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

plane serves as the operational layer of the zero-trust architecture, implementing access control policies and ensuring the security of applications and data. It is responsible for validating and auditing all network requests and actions, encompassing users, devices, applications and data.

In the IIoT era, authentication serves as the primary critical task for ensuring security within the zero-trust framework. However, existing authentication schemes face three critical challenges: (1) Traditional methods often impose high computational overhead on resource-constrained IIoT devices, compromising efficiency; (2) Most solutions lack mechanisms for continuous verification of device legitimacy during active sessions, leaving systems vulnerable to runtime attacks; (3) User authentication in IIoT environments frequently relies on single-factor methods, insufficient for zero-trust requirements. Considering that connected devices in the network must remain secure at all times, authentication should function as a continuous process. Consequently, a continuous authentication scheme is implemented within the zero-trust IIoT framework. The proposed continuous authentication scheme ensures ongoing verification of devices, maintaining consistency between entities and their claimed identities throughout the session on a temporal scale. Entities originating from untrusted spaces must undergo identity authentication and session key negotiation upon initiating authentication requests and starting a session to secure subsequent communications. Specifically, the primary contributions of this study are as follows:

1. To minimize computational overhead on IIoT, the study adopts physical unclonable functions (PUF) for lightweight device authentication. Furthermore, to satisfy continuous authentication requirements in zero-trust IIoT and maintain identity consistency during both initial authentication and the session, location information is utilized for continuous device verification, ensuring session security.
2. A secure signature-based authentication and key agreement protocol is introduced for zero-trust IIoT, incorporating three-factor user identity verification to ensure secure user access. Simultaneously, secure and efficient session keys are generated for users, IIoT devices and gateways to facilitate subsequent communication.
3. The proposed scheme is formally analyzed for security using the BAN logic model, which confirms its robustness. Additionally, comprehensive informal security analysis validates that the scheme withstands known attacks and fulfills design objectives. Performance evaluation indicates that the scheme offers enhanced computational efficiency, notable communication efficiency and extended functionality.

The remaining sections of this article are organized as follows: [Section 2](#) discusses related work, [Section 3](#) provides an introduction, and [Section 4](#) outlines the scheme. Security and performance analyses are presented in [Sections 5 and 6](#), respectively. Finally, [Section 7](#) concludes the article.

2. Related work

Authentication and session key negotiation protocols are fundamental to ensuring reliable communication in IIoT environments [8–11]. Many scholars have proposed various schemes to guarantee the security of authentication and session keys, but most of these schemes still have security vulnerabilities. Ryu et al. [8] discovered weaknesses in Advavoudi et al.'s [12] protocol, such as insider attacks. Kumar and Saxena [13] presented a lightweight authentication protocol free from bilinear pairing. Lohachab et al. [14] presented an ECC-based authentication and key exchange protocol aimed at secure inter-device communication in smart homes, though it lacks perfect forward secrecy. Saeed et al. [15] introduced a lightweight authentication and key exchange protocol enabling secure communication between IoT devices

and cloud servers, supporting secure data transfer between local clients and remote clouds while ensuring data integrity. However, Mitchell et al. [16] showed that Saeed et al.'s [15] scheme is susceptible to impersonation attacks, known session-specific temporary information attacks, and replay attacks. Kumar et al. [17] introduced a signature-based authentication and key exchange protocol tailored for IoT environments. This scheme's key advantage lies in providing user anonymity while maintaining strong resistance to numerous existing attacks. However, the scheme lacks mutual authentication support and remains susceptible to information disclosure attacks. Wang et al. [18] introduced an ECC-based enhanced anonymous authentication scheme designed for smart healthcare systems. Hayouni [19] presented a lightweight authentication protocol tailored for IoT-based systems. Sadhukhan et al. [20] presented a symmetric encryption and ECC-based authentication scheme for users and smart devices. However, this scheme is susceptible to user impersonation, smart card loss, and privileged insider attacks while lacking support for anonymity and dynamic node addition. Shuai et al. [21] introduced an authentication scheme utilizing the Rabin public key cryptosystem. However, this scheme is vulnerable to user impersonation, smart card loss, privileged insider, and offline guessing attacks. Lyu et al. [22] proposed an elliptic curve cryptography-based session key negotiation scheme for smart home environments, incurring relatively high overall costs, and server nodes are not directly involved in session key negotiation. Liu et al. [23] introduced a lightweight three-factor authentication scheme that supports password and biometric updates for users. Tanveer et al. [24] achieved three-factor user authentication, employing hash functions and XOR operations to establish session keys while supporting user revocation. In the same year, Dai and Xu [25] discovered single-point failure vulnerabilities in Guo et al.'s scheme and introduced a novel ECC-based three-factor authentication scheme supporting smart card revocation, dynamic sensor node addition, and single-point failure mitigation. Zhao et al. [26] developed a novel ECC-based three-factor authentication and key exchange protocol tailored for IoT environments, applicable to single-gateway scenarios and scalable to multi-gateway setups. Miao et al. [27] designed a lightweight protocol that effectively reduces communication and computational overhead. Chen et al. [28] introduced a two-factor multi-gateway authentication protocol leveraging passwords and smart cards, designed to resist password guessing, identity guessing, and smart card loss attacks.

As a lightweight physical cryptographic primitive, Physical Unclonable Functions (PUF) possess characteristics that render them applicable in authentication schemes [29–33]. Gassend et al. [34] were the first to apply PUFs to authentication schemes. Shah et al. [35] introduced a PUF-based authentication protocol incorporating a challenge-response mechanism and utilizing the Advanced Encryption Standard (AES) to enhance protocol security. In 2019, an RFID authentication protocol [36] was introduced, addressing unclonability and traceability security requirements and enabling mutual authentication. Reference [37] presented a PUF-based authentication protocol leveraging dynamic identities to prevent real identity exposure and enhance device security. Reference [38] introduced a PUF-based bidirectional authentication protocol supporting authentication between sensors and servers as well as inter-device authentication. Aman et al. [39] proposed an authentication scheme for IIoT supporting both end-to-end and end-to-server authentication. Banerjee et al. [40] identified temporary key leakage vulnerabilities in Gope et al.'s [41] scheme and introduced a secure and efficient protocol supporting mutual authentication between users, gateways, and sensor nodes. Chatterjee et al. [42] identified denial-of-service and synchronization vulnerabilities in [39] and proposed an innovative decentralized PUF-based authentication scheme. This approach avoided the need for verifiers to store PUF challenge-response pairs in a database, effectively mitigating machine learning-based and desynchronization attacks targeting PUFs. Mall et al. [43] reviewed PUF-based schemes within the IIoT context, emphasizing the inherent advantages of PUFs, such as unclonability,

uniqueness and unpredictability, in achieving lightweight and efficient authentication. Although existing authentication schemes provide important references for the Zero Trust IIoT scenario, they still have some limitations. For example, there are deficiencies in computational efficiency, making it difficult to meet the real-time demands of industrial applications. While these schemes support end-to-end authentication, they lack mechanisms for session continuity verification.

3. Preliminaries

3.1. System model

Our system comprises three entities: users, gateways, and IIoT devices. The network model relationships are depicted in Fig. 1. The following provides a detailed description of these entities:

Users: Users initiate communication with the gateway to access IIoT devices. IIoT services are accessible only to registered users through gateway communication.

Gateway (GW): All service users and IIoT devices are required to register at the gateway. Acting as a trusted entity, the gateway handles authentication requests between users and sensor devices.

IIoT devices: Installed within factories, these devices monitor and collect data. Users access sensor devices through the gateway to obtain sensor data. Each sensor in the factory registers securely with the GW, establishing itself as a legitimate node.

3.2. Physical unclonable functions

PUF is a hardware primitive based on a challenge-response mechanism, where it receives a challenge (Challenge, C) and generates the corresponding response (Response, R) from C and the physical characteristics of its integrated chip (Integrated Chip, IC), offering anti-counterfeiting properties for a given device at the hardware level [38–40]. The challenge C received by PUF and the corresponding response R are both in the form of bit strings, making PUF essentially a one-way function $R = \text{PUF}(C)$. The user does not know the internal parameters of PUF, and its security depends on the difficulty of measuring or estimating these parameters and the difficulty of manufacturing two chips with the same set of parameters. Essentially, the security of PUF is based on the fact that even when the manufacturing process for different ICs is the same, each IC typically differs slightly due to manufacturing variations. The basic structure of the arbiter PUF is shown in Fig. 2.

Specifically, PUF has the following functionalities:

- 1) It is impossible to create two identical PUF replicas
- 2) In real-world circuits, the difference between input-output function mappings is fixed and unpredictable. In this sense, it is akin to a hardware-based one-way function.
- 3) Any attempt to tamper with a device containing PUF will change its behavior, thus compromising the PUF.

4. Proposed scheme

Based on [23–44], we propose a scheme. The proposed scheme consists of the initialization phase, user and device registration phase, login and key agreement phase, and device continuous authentication

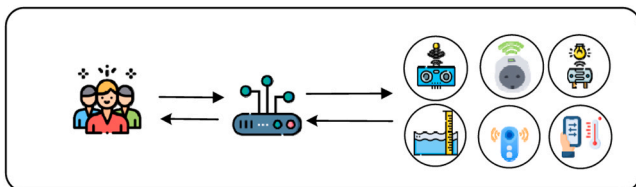


Fig. 1. System model.

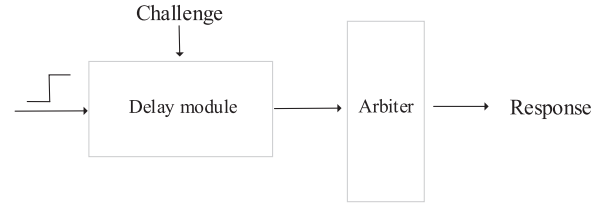


Fig. 2. Basic structure of the arbiter PUF.

phase. The symbols and their corresponding meanings used in the scheme described in this chapter are listed in Table 1.

4.1. Initialization

For system initialization, GW selects a hash function $H(\cdot)$. Additionally, GW chooses an elliptic curve E defined over a finite field F_p . GW then selects a subgroup G of E with a large prime order n , where P serves as the generator point. GW subsequently generates a private key s and derives corresponding $P_{pub} = sP$, where $s \in Z_n^*$. Finally, GW securely stores s while publicly sharing the parameters $\{E(F_p), G, P, P_{pub}\}$.

4.2. Register

(1) User registration

To become a legitimate participant in the system, U_i and GW execute the following steps, enabling U_i to access sensor data. The detailed process is illustrated in Fig. 3.

Step 1: U_i provides identity ID_i , password PW_i and biometric information O_i , chooses random number α_i and calculates $Gen(O_i) = (\tau_i, \theta_i)$. U_i then computes $HID_i = H(\alpha_i, ID_i)$, $PID_i = H(ID_i, PW_i, \tau_i, \alpha_i)$, submitting $\{ID_i, PID_i, HID_i\}$ to GW for registration.

Step 2: Upon receiving the registration request, GW verifies the presence of ID_i in its database. If found, U_i must provide a new identity. Otherwise, GW generates a random number g_i , assigns an anonymous identity FID_i , computes $PK_{i1} = g_iP$, $FHP_i = H(FID_i, PK_{i1}, P_{pub})$, $psk_i = g_i + sFHP_i$, $H_i = H(PID_i, FID_i, psk_i, PK_{i1})$. The values $\{psk_i, PK_{i1}, H_i\}$ are stored in the smart card SC_i and provided to U_i .

Step 3: After receiving SC_i , U_i calculates $Y_i = \alpha_iP$, $FPX_i = H(FID_i, PID_i, HID_i, Y_i)$, $PK_{i2} = PK_{i1} + FPX_iY_i$, $A_i = H(ID_i, PW_i, \tau_i) \oplus \alpha_i$, $B_i = H(ID_i, PW_i, \alpha_i) \oplus psk_i$. U_i stores $\{A_i, \theta_i, PK_{i2}\}$ in SC_i , replaces psk_i with B_i , and announces FID_i and $PK_i = (PK_{i1}, PK_{i2})$.

(2) Device Registration

Sensor device SD_j utilizes PUF functionality for registration, registering its information with GW. The detailed process is illustrated in

Table 1
Symbols.

Symbols	Meaning
U_i	User
GW	Gateway
s	Secret key of GW
P_{pub}	Public key of GW
SD_j	j-th device
SK	Session key
T_i	Timestamps
ID_i	Identity of U_i
PW_i	Password of U_i
O_i	biometric information of U_i
$Gen()/Rep()$	Generation and reproduction algorithm of fuzzy extractor
PUF	Physical Unclonable Function
$\oplus$	Bitwise XOR operator
$H()$	One-way hash function

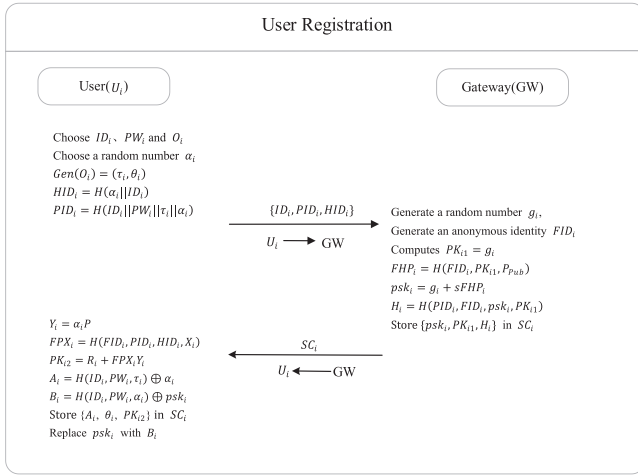


Fig. 3. User registration.

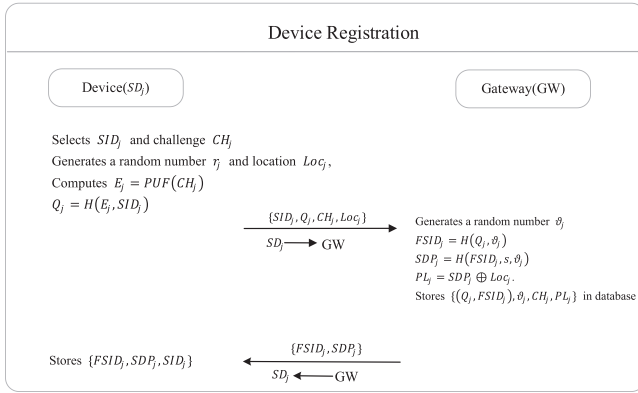


Fig. 4. Device registration.

Fig. 4, with the steps outlined as follows:

Step 1: SD_j selects SID_j and PUF challenge CH_j . It generates a random number r_j and location information Loc_j , then computes $E_j = PUF(CH_j)$, $Q_j = H(E_j, SID_j)$. Finally, SD_j sends $\{SID_j, Q_j, CH_j, Loc_j\}$ to GW.

Step 2: GW verifies that SID_j is stored in its database. If SID_j in the database, GW generates a random number θ_j and calculates $FSID_j = H(Q_j, \theta_j)$, $SDP_j = H(FSID_j, s, \theta_j)$, $PL_j = SDP_j \oplus Loc_j$. GW stores $\{(Q_j, FSID_j), \theta_j, CH_j, PL_j\}$ in its database and securely transmits $\{FSID_j, SDP_j\}$ to SD_j .

Step 3: Upon receiving the message, SD_j saves $\{FSID_j, SDP_j, SID_j\}$.

4.3. Login and authentication phase

After logging in with their smart card and credentials, U_i sends an authentication request to GW. Upon confirmation, GW forwards an authentication message to the corresponding SD_j . Each entity validates the response. Afterward, U_i , GW and SD_j establish a session key to guarantee secure communication. The detailed process is shown in Fig. 5 and outlined as follows:

Step 1: U_i inputs ID_i , PW_i and O_i . SC computes $Rep(O_i, \theta_i) = \tau_i, \alpha_i = H(ID_i, PW_i, \tau_i) \oplus A_i$, $psk_i = H(ID_i, PW_i, \alpha_i) \oplus B_i$, $PID_i = H(ID_i, PW_i, \tau_i, \alpha_i)$, $HID_i = H(\alpha_i, ID_i)$ and $H_i = H(PID_i, HID_i, psk_i, PK_{i1})$. If $H_i = H_i$ proceed; otherwise, end the session.

Step 2: After logging in, user generates a message and signature. They choose two random numbers u_i, r_i and timestamp T_1 , calculate $\mathcal{Z}_i = u_i P$, $E_i = u_i P_{pub}$, $m_i = E_i \oplus (HID_i || r_i)$, $FHP_i = H(FID_i, PK_{i1}, P_{pub})$, $FPT_i = H(FID_i, PK_{i1}, PK_{i2}, m_i, \mathcal{Z}_i, T_1)$ and create the signature $\sigma_i = (\mathcal{Z}_i, S_i)$,

where $S_i = u_i + FPT_i(psk_i + FHP_i \alpha_i)$. The user sends $M_1 = \{m_i, \sigma_i, T_1, FID_i, FSID_j\}$ to GW.

Step 3: GW ensures the freshness of T_1 and validates the signature $S_i P = U_i + FPT_i(PK_{i2} + FHP_i P_{pub})$ by computing $FHP_i = H(FID_i, PK_{i1}, P_{pub})$ and $FPT_i = H(FID_i, PK_{i1}, PK_{i2}, m_i, \mathcal{Z}_i, T_1)$. If valid, GW calculates $E_i = s \mathcal{Z}_i$ to derive $HID_i || r_i = E_i \oplus m_i$, retrieves θ_j and CH_j for $FSID_j$, generates N_g and T_2 , and computes $SDP_j = H(FSID_j, H(s, \theta_j))$, $P_1 = CH_j \oplus H(FSID_j, SDP_j)$, $P_2 = H(r_i, HID_i) || N_g$, $P_3 = P_2 \oplus H(Q_j, CH_j, SDP_j)$ and $S_j = H(P_2, T_2, Q_j, CH_j, SDP_j)$. GW transmits $M_2 = \{S_j, P_1, P_3, T_2\}$ to SD_j over a wireless channel.

Step 4: Upon receiving M_2 , SD_j verifies the freshness of T_2 . It calculates $CH_j = P_1 \oplus H(FSID_j, SDP_j)$, $PUF(CH_j) = E_j$, $Q_j = H(SID_j, E_j)$, $H(r_i, HID_i) || N_g = P_3 \oplus H(Q_j, CH_j, SDP_j)$, and $S'_j = H(P_2, T_2, Q_j, CH_j, SDP_j)$. If $S'_j = S_j$, it generates N_j and timestamp T_3 , computes $SK = H(N_j, H(r_i, HID_i), N_g, H(SDP_j))$, $P_4 = N_j \oplus H(FSID_j, CH_j, SDP_j, T_3)$, $SP_j = H(CH_j, SK, Q_j)$ and transmits $M_3 = \{SP_j, P_4, T_3\}$ to GW over a wireless channel.

Step 5: After receiving M_3 , GW verifies T_3 for freshness and computes $N_j = P_4 \oplus H(FSID_j, CH_j, SDP_j, T_3)$, $SK = H(N_j, H(r_i, HID_i), N_g, H(SDP_j))$ and $SP'_j = H(CH_j, SK, Q_j)$. It checks if $SP'_j = SP_j$. Upon validation, GW generates a random a_g , computes $G_j = a_g P$, $L_j = a_g U_i$, $P_5 = SK \oplus H(L_j, HID_i, u_i)$ and $SU_j = H(SK, P_5, L_j, G_j, u_i)$ and transmits $M_4 = \{G_j, P_5, SU_j\}$ to U_i .

Step 6: Upon receiving M_4 , U_i computes $L_j = u_i G_j$, $SK = P_5 \oplus H(L_j, HID_i, r_i)$ and $SU'_j = H(SK, P_5, L_j, G_j, r_i)$. If $SU'_j = SU_j$, SK is accepted.

4.4. Continuous authentication

In the Zero Trust IIoT, connected devices must remain in a secure state at all times, and authentication should be an ongoing process. Therefore, in this phase, location information from the registration process is utilized for continuous authentication of the device, ensuring its security throughout the session.

Step 1: The device generates current location Loc'_j and timestamp T_4 .

It computes $Lp_j = Loc'_j \oplus SDP_j$, $G_j = Lp_j \oplus H(SK)$ and $V_j = H(Lp_j, G_j, T_4)$. It then sends $M_5 = \{V_j, T_4, FSID_j\}$ to GW.

Step 2: After receiving M_5 , GW verifies the freshness of T_4 . It retrieves PL_j for $FSID_j$, computes $G'_j = PL_j \oplus H(SK)$ and $V'_j = H(PL_j, G_j, T_4)$. It checks if $V'_j = V_j$. If not equal, the session is terminated.

5. Security evaluation

5.1. Security proof

BAN logic is used in this article to analyze the correctness of the scheme. The following are the five rules applied:

RULE (1)- Message meaning rule:

$$\frac{M \equiv M \xrightarrow{K} N, M \triangleleft \{X\}_K}{M \equiv N \sim X}$$

RULE (2)- Nonce verification rule:

$$\frac{M \#(X), M \equiv N \sim X}{M \equiv N \equiv X}$$

RULE (3)- Jurisdiction rule:

$$\frac{M \equiv N \Rightarrow X, M \equiv N \equiv X}{M \equiv X}$$

RULE (4)- Belief rule:



Fig. 5. Login and authentication phase.

$$\frac{M \equiv X, M \equiv Y}{M \equiv (X, Y)}$$

RULE (5)- Freshness rule:

$$\frac{M \equiv \#(X)}{M \equiv \#(X, Y)}$$

The following objectives must be fulfilled:

$$\text{Goal1} : U_i \equiv (U_i \longleftrightarrow^{SK} GW)$$

$$\text{Goal2} : U_i \equiv GW \equiv (U_i \longleftrightarrow^{SK} GW)$$

$$\text{Goal3} : GW \equiv (U_i \longleftrightarrow^{SK} GW)$$

$$\text{Goal4} : GW \equiv U_i \equiv (U_i \longleftrightarrow^{SK} GW)$$

$$\text{Goal5} : SD_j \equiv (SD_j \longleftrightarrow^{SK} GW)$$

$$\text{Goal6} : SD_j \equiv GW \equiv (SD_j \longleftrightarrow^{SK} GW)$$

$$\text{Goal 7} : GW \equiv (SD_j \longleftrightarrow^{SK} GW)$$

$$\text{Goal 8} : GW \equiv SD_j \equiv (SD_j \longleftrightarrow^{SK} GW)$$

Protocol idealization

$$M1 : U_i \rightarrow GW : \{HID_i, u_i, T_1\}_{E_i}$$

$$M2 : GW \rightarrow SD_j : \{H(u_i, HID_i), N_g, T_2\}_{SDP_j}$$

$$M3 : SD_j \rightarrow GW : \{H(u_i, HID_i), N_j, N_g, T_3, SDP_j\}_{SDP_j}$$

$$M4 : GW \rightarrow U_i : \{H(N_j, H(u_i, HID_i), N_g, H(SDP_j))\}_{L_j}$$

The assumptions are as follow:

$$A_1 : GW \equiv U_i \xleftrightarrow{E_i} GW$$

$$A_2 : GW \equiv \#(T_1)$$

$$A_3 : SD_j \equiv GW \xleftrightarrow{SDP_j} SD_j$$

$$A_4 : SD_j \equiv \#(T_2)$$

$$A_5 : GW \equiv GW \xleftrightarrow{SDP_j} SD_j$$

$$A_6 : GW \equiv \#(T_3)$$

$$A_7 : U_i \equiv U_i \xleftrightarrow{L_j} GW$$

$$A_8 : U_i \equiv \#(N_g)$$

$$A_9 : U_i \equiv GW \Rightarrow (U_i \xleftrightarrow{SK} GW)$$

$$A_{10} : GW \equiv U_i \mid \Rightarrow (U_i \xleftrightarrow{SK} GW)$$

$$A_{11} : SD_j \mid \equiv GW \mid \Rightarrow (SD_j \xleftrightarrow{SK} GW)$$

$$A_{12} : GW \mid \equiv SD_j \mid \Rightarrow (SD_j \xleftrightarrow{SK} GW)$$

Proof:

Based on M1, we get:

$$R_1 : GW \triangleleft \{HID_i, u_i, T_1\}_{E_i}$$

From R_1, A_1 and message meaning rule, it gets:

$$R_2 : GW \mid \equiv U_i \mid \sim \{HID_i, u_i, T_1\}_{E_i}$$

Based on R_2, A_2 , and the freshness rule, it obtains:

$$R_3 : GW \mid \equiv \#(HID_i, u_i, T_1)$$

Based on R_3, R_2 , and apply the Nonce verification rule, we get:

$$R_4 : GW \mid \equiv U_i \mid \equiv (HID_i, u_i, T_1)$$

Based on M2, it gets:

$$R_5 : SD_j \triangleleft \{H(u_i, HID_i), N_g, T_2\}_{SDP_j}$$

From R_5, A_3 and message meaning rule, we get:

$$R_6 : SD_j \mid \equiv GW \mid \sim \{H(u_i, HID_i), N_g, T_2\}_{SDP_j}$$

Based on R_6, A_4 , and Freshness rule, we obtain:

$$R_7 : SD_j \mid \equiv \#(H(u_i, HID_i), N_g, T_2)$$

According to R_7, R_6 , and Nonce verification rule, we get:

$$R_8 : SD_j \mid \equiv GW \mid \equiv (H(u_i, HID_i), N_g, T_2)$$

According to M3, it gets:

$$R_9 : GW \triangleleft \{H(u_i, HID_i), N_j, N_g, T_3, SDP_j\}_{SDP_j}$$

From R_9, A_5 and message meaning rule, it gets:

$$R_{10} : GW \mid \equiv SD_j \mid \sim \{H(u_i, HID_i), N_j, N_g, T_3, SDP_j\}_{SDP_j}$$

From R_{10}, A_6 , and freshness rule, we obtain:

$$R_{11} : GW \mid \equiv \#(H(u_i, HID_i), N_j, N_g, T_3, SDP_j)$$

Based on R_{11}, R_{10} and Nonce verification rule, we get:

$$R_{12} : GW \mid \equiv SD_j \mid \equiv (H(u_i, HID_i), N_j, N_g, T_3, SDP_j)$$

According to M4, it gets:

$$R_{13} : U_i \triangleleft \{H(N_j, H(u_i, HID_i), N_g, H(SDP_j))\}_{L_j}$$

According to R_{13}, A_7 , and Message meaning rule, we get:

$$R_{14} : U_i \mid \equiv GW \mid \sim \{H(N_j, H(u_i, HID_i), N_g, H(SDP_j))\}_{L_j}$$

According to R_{14}, A_8 , and Freshness rule, we get:

$$R_{15} : U_i \mid \equiv \#(H(N_j, H(u_i, HID_i), N_g, H(SDP_j)))$$

From R_{15}, R_{14} and Nonce verification rule we can get:

$$R_{16} : U_i \mid \equiv GW \mid \equiv (H(N_j, H(u_i, HID_i), N_g, H(SDP_j)))$$

Based on R_8, A_{12} , we obtain:

$$R_{17} : SD_j \mid \equiv |GW \mid \equiv (SD_j \xleftrightarrow{SK} GW) \text{ (Goal6)}$$

$$R_{18} : GW \mid \equiv SD_j \mid \equiv (SD_j \xleftrightarrow{SK} GW) \text{ (Goal8)}$$

Based on $R_{17}, R_{18}, A_{11}, A_{12}$ and Jurisdiction rule, we obtain:

$$R_{19} : SD_j \mid \equiv (SD_j \xleftrightarrow{SK} GW) \text{ (Goal5)}$$

$$R_{20} : GW \mid \equiv (SD_j \xleftrightarrow{SK} GW) \text{ (Goal7)}$$

Based on R_4, R_{16} and Jurisdiction rule, we obtain:

$$R_{21} : U_1 \mid \equiv GW \mid \equiv (U_1 \xleftrightarrow{SK} GW) \text{ (Goal2)}$$

$$R_{22} : GW \mid \equiv |U_1 \mid \equiv (U_1 \xleftrightarrow{SK} GW) \text{ (Goal4)}$$

Based on $R_{21}, R_{22}, A_9, A_{10}$ and Jurisdiction rule, we obtain:

$$R_{23} : U_i \equiv (U_i \xrightarrow{SK} GW) \text{ (Goal1)}$$

$$R_{24} : U_i \equiv (U_i \xrightarrow{SK} GW) \text{ (Goal3)}$$

From the above, we can see that four goals have been proved.

5.2. Security analysis

(1) Mutual authentication

In the proposed scheme, entities register through gateway. During login and key agreement, the gateway authenticates user identities by verifying the signature in M_1 . Similarly, only the gateway possessing the key s can derive $(HID_i || u_i)$, to generate the key. The GW validates the device using PUF technology, ensuring that only a gateway possessing the secret value SDP_j can retrieve the session key SK. Thus, all entities mutually authenticate by validating the messages M_1, M_2, M_3 and M_4 . Moreover, these messages dynamically change based on random numbers and T_i . Hence, the proposed scheme guarantees mutual authentication.

(2) Temporary Secret Leakage Attack

Suppose the random numbers generated during the session are leaked. $\mathcal{A}$ attempts to compute SK using these values but cannot calculate $SK = H(N_j, H(u_i, HID_i), N_g, H(SDP_j))$. To compute a valid SK, knowledge of HID_i and SDP_j is required. Without the gateway key s , $\mathcal{A}$ cannot retrieve HID_i from the messages or access SDP_j . Thus, the proposed scheme effectively resists session-specific random number leakage attacks.

(3) Perfect Forward Secrecy

In this scheme, SK is negotiated and generated using secret values N_j, u_i , and N_g , which are encrypted during authentication. Consequently, no adversary $\mathcal{A}$ can access these secret values. Additionally, even if an adversary somehow obtains the current session key, the updated values of N_j, u_i and N_g prevent the retrieval of previous session keys. Thus, this scheme ensures perfect forward secrecy.

(4) Replay Attack

In this protocol, timestamps are included in messages sent from the user to the gateway and from the gateway to the sensor to ensure integrity. The recipient verifies the freshness of the timestamp upon receiving the message. If the received timestamp is outdated, the recipient denies the request and ends the session. Similarly, when the sensor sends messages to the gateway and the gateway sends messages to the user, random numbers ensure session freshness. Thus, the scheme is resistant to replay attacks.

(5) Privileged Insider Attack

The privileged user of the GW (potentially an adversary $\mathcal{A}$) may access $\{ID_i, PID_i, HID_i\}$, the registration data of U_i during the registration phase. Suppose U_i 's smart card SC_i is lost or stolen by $\mathcal{A}$ after registration. Even if $\mathcal{A}$ extracts all stored data from SC_i , $\mathcal{A}$ cannot infer ID_i or PW_i . ID_i is not directly stored on SC_i . Additionally, the values psk_i and α_i stored in SC_i are secured by the one-way hash function $H(\cdot)$. For $\mathcal{A}$ to correctly guess ID_i and PW_i , they must also know the biometric key τ_i and the secret values (psk_i, α_i) . To extract psk_i and α_i , $\mathcal{A}$ would require knowledge of ID_i and PW_i , since $A_i = H(ID_i, PW_i, \tau_i) \oplus \alpha_i$ and $B_i = H(ID_i, PW_i, \alpha_i) \oplus psk_i$. Thus, the proposed scheme securely resists this type of attack.

(6) Denial-of-Service Attacks

The proposed scheme utilizes three-factor authentication, where U_i must provide the correct ID_i , PW_i , and O_i to initiate authentication and key agreement requests. Moreover, all messages incorporate timestamps, ensuring that only fresh messages are accepted. Thus, the proposed scheme effectively resists denial-of-service attacks.

(7) Stolen Smart Card Attack

If U_i 's smart card SC_i is lost or stolen, an adversary $\mathcal{A}$ may gain access to all stored data. However, U_i 's secret credentials remain secure. The secret values (psk_i, α_i) of U_i are XORed with ID_i, PW_i , and τ_i , and secured using a one-way hash function. Thus, the proposed scheme effectively resists this attack.

(8) User Impersonation Attack

Assume an adversary $\mathcal{A}$ intercepts U_i 's login request message $\{ID_i, PID_i, HID_i\}$ and tries to forge a valid login request by impersonating U_i . To achieve this, $\mathcal{A}$ selects a random number u_i and attempts to compute $U_i = u_i P$, $E_i = u_i P_{Pub}$, $m_i = E_i \oplus (HID_i || u_i)$, $\sigma_i = (U_i, S_i)$, $FHP_i = H(FID_i, PK_{i1}, P_{Pub})$, $FPT_i = H(FID_i, PK_{i1}, PK_{i2}, m_i, \mathcal{Z}_i, T_1)$ and $S_i P = U_i + FPT_i(PK_{i2} + FHP_i P_{Pub})$. Here, $\mathcal{A}$ requires knowledge of U_i 's private keys (psk_i, α_i) to compute $S_i P = U_i + FPT_i(PK_{i2} + FHP_i P_{Pub})$. Consequently, recreating the login request through eavesdropping is infeasible, making our scheme resistant to such attacks.

(9) Man-in-the-Middle Attack

Suppose an adversary $\mathcal{A}$ intercepts the login request message $M_1 = \{m_i, \sigma_i, T_1, FID_i, FSID_j\}$ and tries to modify it into another valid login request. To achieve this, $\mathcal{A}$ could select a random number u_i and generate the current timestamp T_1 . Then, $\mathcal{A}$ might compute $U_i = u_i P$, $E_i = u_i P_{Pub}$, $m_i = E_i \oplus (HID_i || u_i)$, $\sigma_i = (U_i, S_i)$, $FHP_i = H(FID_i, PK_{i1}, P_{Pub})$, $FPT_i = H(FID_i, PK_{i1}, PK_{i2}, m_i, \mathcal{Z}_i, T_1)$ and $S_i P = U_i + FPT_i(PK_{i2} + FHP_i P_{Pub})$. However, without (psk_i, α_i) , $\mathcal{A}$ would find it infeasible to compute $S_i P = U_i + FPT_i(PK_{i2} + FHP_i P_{Pub})$. Thus, $\mathcal{A}$ cannot forge a valid login request message. Similarly, $\mathcal{A}$ is incapable of creating other valid messages during the authentication and key establishment phases. Hence, the proposed scheme resists man-in-the-middle attacks.

(10) Anonymity and Untraceability

In the key agreement phase, the user employs the anonymous identity FID_i to interact with the gateway. Assume $\mathcal{A}$ intercepts $M_1 = \{m_i, \sigma_i, T_1, FID_i, FSID_j\}$, $M_2 = \{S_j, P_1, P_3, T_2\}$, $M_3 = \{SP_j, P_4, T_3\}$ and $M_4 = \{G_j, P_5, SU_j\}$ during login and authentication. Because random numbers and current timestamps are utilized,

M_1, M_2, M_3 , and M_4 are dynamid and unique for each session. Additionally, none of these messages explicitly include plaintext identities ID_i and SID_j during insecure channel transmissions. Hence, the proposed scheme ensures both anonymity and untraceability.

6. Performance analysis

6.1. Security features comparison

This compares the proposed scheme with related existing schemes regarding security features. In Table 2, $\checkmark$ indicates the feature is met, and $\times$ indicates it is not.

6.2. Computation overhead

In this section, we compare the computational and communication overhead of the proposed scheme with other related schemes. Since the frequency of execution in the registration stages is low, we only consider analyzing the login and mutual authentication stages. The execution time of XOR operations is negligible, so we exclude it from the computational overhead. In accordance with [3], Table 3 summarizes the approximate time required for each cryptographic operation.

From the data in the Table 4 and Fig. 6, the proposed scheme shows significant optimization in computational overhead on the user, gateway, and device sides. On the user side, the proposed scheme has a computation overhead of

$3T_e + 9T_h + T_b$, which is a significant reduction compared to [45] [46] [48], as it reduces the number of ECC point multiplications. Therefore, the computational overhead on the user side is greatly reduced. On the device side, the scheme uses PUF and does not utilize ECC point multiplication, thus reducing the computation overhead on the device side compared to other schemes.

Overall, the proposed scheme demonstrates significant optimization across all aspects, particularly in reducing computational overhead and maintaining moderate computation time, which enhances the overall efficiency of the system. Compared to traditional schemes, the proposed scheme reduces computational overhead on both the user and gateway sides, while also significantly optimizing computation time. This makes the proposed scheme highly applicable in multi-terminal distributed computing environments, especially in resource-constrained scenarios, with great application potential.

6.3. Communication overhead

The communication consumption of all proposals is shown in Table 5. To estimate the communication costs, we assume the timestamp

Table 2
Security features comparison.

Feature	[45]	[46]	[47]	[48]	Our
F1	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
F2	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
F3	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
F4	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
F5	$\checkmark$	$\checkmark$	$\checkmark$	$\times$	$\checkmark$
F6	$\times$	$\times$	$\times$	$\checkmark$	$\checkmark$
F7	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
F8	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
F9	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
F10	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
F11	$\checkmark$	$\checkmark$	$\times$	$\checkmark$	$\checkmark$

F1: Mutual authentication; F2: Temporary Secret Leakage Attack; F3: Perfect Forward Secrecy; F4: Replay Attack; F5: Privileged Insider Attack; F6: Denial-of-Service Attacks; F7: Stolen Smart Card Attack; F8: User Impersonation Attack; F9: Man-in-the-Middle Attack; F10: Anonymity and Untraceability; F11: Three-Factor authentication.

Table 3
Operation cost time.

Symbol	Description	Time(ms)
T_e	ECC point multiplication	4.12
T_h	Hash function	0.311
T_a	ECC point addition	0.273
T_b	Fuzzy extraction	4.12
T_f	PUF function	0.51μ
T_s	Symmetric cryptography operation	0.413

Table 4
Computation overhead.

Scheme	User	Gateway	Device	Time (ms)
[45]	$16T_h + 5T_e + 2T_a + T_b$	$9T_h + 3T_e + 2T_a$	$8T_h + 4T_e + 1T_a$	65.188
[46]	$19T_h + 4T_e + T_a + T_b$	$T_h + 5T_e + T_a$	$12T_h + 4T_e + T_a$	68.451
[47]	$2T_e + 4T_h$	$5T_h + 2T_e + 2T_a$	$3T_h + T_e + 2T_a$	33.664
[48]	$7T_h + 4T_e + T_a$	$4T_h + T_e + T_a$	$6T_h + 4T_e + 2T_a$	47.579
Our	$3T_e + 9T_h + T_b$	$3T_e + 14T_h + T_a$	$7T_h + T_f$	38.443

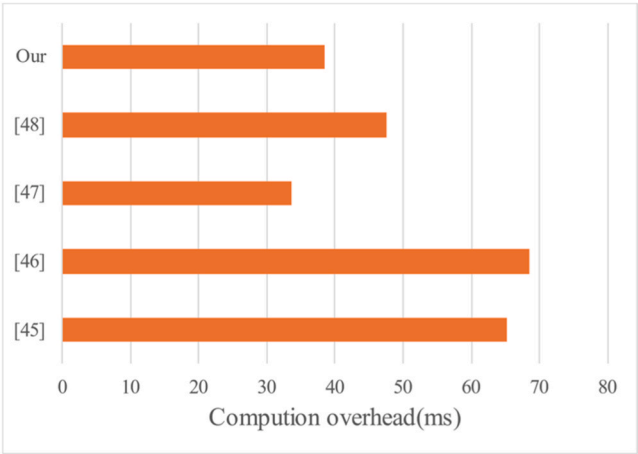


Fig. 6. Computation overhead.

Table 5
Communication overhead.

Scheme	Communication Overhead(bits)
[45]	3200
[46]	3360
[47]	2560
[48]	3104
Our	2688

size is 32 bits, the random number size is 128 bits, the size of all identities is 32 bits, the hash output size is 160 bits, and the ECC point size is 320 bits.

According to Table 5 and Fig. 7, the communication overhead of the proposed scheme is 2688 bits, demonstrating relatively superior performance compared to all other schemes. Compared to other schemes, the communication overhead of the proposed scheme is notably lower than some high-cost schemes. Compared to the 3200 bits of [45] scheme, the communication overhead of the proposed scheme is reduced by 16 %, demonstrating a noticeable optimization effect.

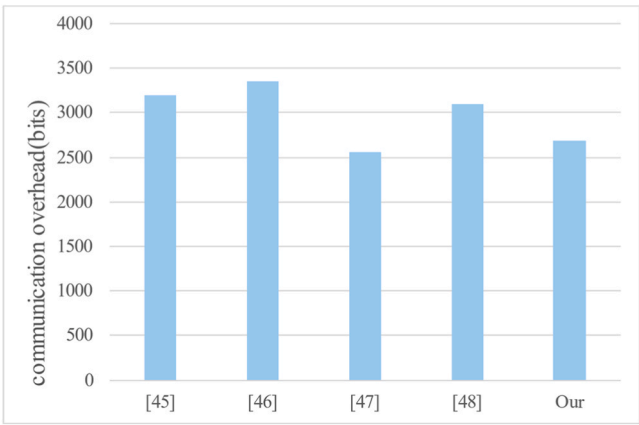


Fig. 7. Communication overhead.

Although the communication overhead of the [46] scheme is 3360 bits, which is 24.9 % higher than the proposed scheme, the difference in communication overhead between the two schemes is substantial, highlighting the good adaptability of the proposed scheme across multiple schemes.

Overall, the communication overhead of the proposed scheme is relatively low, providing advantages in bandwidth saving and resource optimization compared to many traditional schemes. This makes the proposed scheme more practical in bandwidth-limited environments, particularly for scenarios that require efficient communication with restricted bandwidth.

7. Conclusion

With the development of the Industrial Internet of Things (IIoT), the application domain of IoT continues to expand. Consequently, secure authentication schemes are essential to ensure the security of IIoT services. This paper proposes a Zero Trust framework for IIoT and introduces an authentication and key exchange scheme tailored to this framework. On the IIoT device side, PUF technology is utilized for device authentication, while location information is leveraged for continuous authentication to ensure secure device access. Compared to traditional boundary defense schemes, the proposed approach achieves continuous verification through PUF-based device authentication and location-aware session monitoring. While ensuring the security of PUF, the scheme also reduces authentication latency. To address the limitations of device authentication, the proposed scheme integrates three-factor user authentication, effectively mitigating the risks of credential theft associated with single-factor methods. BAN logic is employed to demonstrate that the proposed scheme ensures secure mutual authentication. Furthermore, performance analysis reveals that, compared to other authentication schemes, this approach provides more secure services in IIoT environments. These advantages position the proposed scheme as an efficient practical path for implementing Zero Trust in IIoT scenarios.

Author Contributions

Conceptualization, Tao Wan and Huan Wang; methodology, Tao Wan and Buhai Shi; software, validation, formal analysis, investigation, Tao Wan, Huan Wang and Buhai Shi; writing—original draft preparation, Tao Wan; writing—review and editing, Tao Wan and Huan Wang. All authors have read and agreed to the published version of the manuscript.

CRediT authorship contribution statement

Wan Tao: Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Shi Buhai:** Writing – review & editing, Validation, Software, Methodology, Investigation, Formal analysis. **Wang Huan:** Writing – review & editing, Validation, Software, Investigation, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All relevant data are within the paper.

References

- [1] N. Kumar, S. Challa, M. Wazid, A.K. Das, A.G. Reddy, E.-J. Yoon, K.-Y. Yoo, Secure signature-based authenticated key establishment scheme for future IoT applications, *IEEE Access* 5 (1) (2017) 3028–3043.
- [2] Amar Khelloufi, et al., A multimodal latent-features-based service recommendation system for the social Internet of Things, *IEEE Trans. Comput. Soc. Syst.* (2024).
- [3] M. Tanveer, A. Alkhayyat, N. Kumar, et al., REAP-IoT: resource-efficient authentication protocol for the industrial Internet of Things, *IEEE Internet Things J.* 9 (23) (2022) 24453–24465.
- [4] Abdenacer Naouri, et al., Efficient fog node placement using nature-inspired metaheuristic for IoT applications, *Clust. Comput.* (2024) 1–17.
- [5] S. Son, D. Kwon, S. Lee, et al., A zero-trust authentication scheme with access control for 6G-enabled IoT Environments, *IEEE Access* (2024).
- [6] Abdelkarim Ben Sada, et al., A context-aware edge computing framework for smart internet of things, *Future Internet* 15 (5) (2023) 154.
- [7] Jian Wang, Fan Li*, Song Lv, Lijun He, Chao Shen, Physically realizable adversarial creating attack against vision-based BEV space 3D object detection, *IEEE Trans. Image Process.* 34 (2025) 538–551.
- [8] J. Ryu, D. Kang, H. Lee, et al., A secure and lightweight three-factor-based authentication scheme for smart healthcare systems, *Sensors* 20 (24) (2020) 7136.
- [9] Jian Wang, Fan Li*, Lijun He, A unified framework for adversarial patch attacks against visual 3d object detection in autonomous driving (published online,), *IEEE Trans. Circuits Syst. Video Technol.* (2025).
- [10] J. Huang, X. Yu, D. An, X. Ning, J. Liu, P. Tiwari, Uniformity and deformation: a benchmark for multi-fish real-time tracking in the farming, *Expert Syst. Appl.* (2025).
- [11] L. Zhang, J. Liu, Y. Wei, D. An, X. Ning, Self-supervised learning-based multi-source spectral fusion for fruit quality evaluation: a case study in mango fruit ripeness prediction, *Inf. Fusion* (2025).
- [12] A. Adavoudi-Jolfaei, M. Ashouri-Talouki, S.F. Aghili, Lightweight and anonymous three-factor authentication and access control scheme for real-time applications in wireless sensor networks, *Peer-to-Peer Netw. Appl.* 12 (2019) 43–59.
- [13] M. Kumar, P.C. Saxena, PF-AID-2KAP: Pairing-free authenticated identity-based two-party key agreement protocol for resource-constrained devices. *Futuristic Trends in Network and Communication Technologies: First International Conference, FTNCT 2018, Solan, India, February 9–10, 2018, Revised Selected Papers 1*, Springer Singapore, 2019, pp. 425–440.
- [14] A. Lohachab, ECC based inter-device authentication and authorization scheme using MQTT for IoT networks, *J. Inf. Secur. Appl.* 46 (2019) 1–12.
- [15] M.E.S. Saeed, Q.Y. Liu, G.Y. Tian, et al., AKAIoTs: authenticated key agreement for Internet of Things, *Wirel. Netw.* 25 (6) (2019) 3081–3101.
- [16] Mitchell C.J. The Saeed-Liu-Tian-Gao-Li authenticated key agreement protocol is insecure[J]. *arXiv preprint arXiv:1906.09330*, 2019.
- [17] N. Kumar, M. Wazid, A.K. Das, V. Odelu, M. Conti, M. Jo, Design of secure user authenticated key management protocol for generic IoT networks, *IEEE Internet Things J.* 5 (1) (2017) 269–282.
- [18] W. Yuanbing, L. Wanrong, L. Bin, An improved authentication protocol for smart healthcare system using wireless medical sensor network, *IEEE Access* 9 (2021) 105101–105117.
- [19] H. Hayouni, AuthenIoT: a lightweight authentication protocol for the internet of things based wireless sensor networks, *EAI Endorsed Trans. Cloud Syst.* 7 (21) (2021).
- [20] D. Sadhukhan, S. Ray, G.P. Biswas, et al., A lightweight remote user authentication scheme for IoT communication using elliptic curve cryptography, *J. Supercomput.* 77 (2) (2021) 1114–1151.
- [21] M. Shuai, L. Xiong, C. Wang, et al., A secure authentication scheme with forward secrecy for industrial internet of things using Rabin cryptosystem, *Comput. Commun.* 160 (2020) 215–227.
- [22] Q. Lyu, N. Zheng, H. Liu, et al., Remotely access “my” smart home in private: an antitracking authentication and key agreement scheme, *IEEE Access* 7 (2019) 41835–41851.
- [23] Z. Liu, C. Guo, B. Wang, A physically secure, lightweight three-factor and anonymous user authentication protocol for IoT, *IEEE Access* 8 (2020) 195914–195928.
- [24] M. Tanveer, G. Abbas, Z.H. Abbas, et al., LAKE-6SH: lightweight user authenticated key exchange for 6LoWPAN-based smart homes, *IEEE Internet Things J.* 9 (4) (2022) 2578–2591.
- [25] C. Dai, Z. Xu, A secure three-factor authentication scheme for multigateway wireless sensor networks based on elliptic curve cryptography, *Ad Hoc Netw.* 127 (2022) 102768.
- [26] X. Zhao, D. Li, H. Li, Practical three-factor authentication protocol based on elliptic curve cryptography for industrial internet of things, *Sensors* 22 (19) (2022) 7510.
- [27] J. Miao, Z. Wang, X. Xue, et al., Lightweight and secure D2D Group Communication for Wireless IoT, *Front. Phys.* 11 (2023) 433.
- [28] C. Chen, H. Guo, Y. Wu, et al., A novel two-factor multi-gateway authentication protocol for WSNs, *Ad Hoc Netw.* 141 (2023) 103089.
- [29] N. Xiao, Z. Wang, X. Sun, et al., A novel blockchain-based digital forensics framework for preserving evidence and enabling investigation in industrial Internet of Things, *Alex. Eng. J.* 86 (2024) 631–643.
- [30] H. Wei, J. Miao, J. Lv, C.-M. Chen, et al., Secure and trustworthy data management mechanism for dance-consumer electronics in AIoT, *IEEE Trans. Consum. Electron.* (2024).
- [31] Jinchao Chen, Tingyang Li, Ying Zhang, Tao You, Yantao Lu, Prayag Tiwari, Neeraj Kumar, Global-and-local attention-based reinforcement learning for cooperative behaviour control of multiple UAVs[J], *IEEE Trans. Veh. Technol.* 73 (3) (2024) 4194–4206.
- [32] Ben Sada, et al., Hybrid metaheuristics for selective inference task offloading under time and energy constraints for real-time IoT sensing systems, *Clust. Comput.* (2024) 1–17.
- [33] J. Miao, Z. Wang, M. Wang, S. Garg, M.S. Hossain, J.J. Rodrigues, Secure and efficient communication approaches for Industry 5.0 in edge computing, *Comput. Netw.* (2024) 110244.
- [34] B. Gassend, D. Clarke, M. Van Dijk, et al., Silicon physical random functions, *Proc. 9th ACM Conf. Comput. Commun. Secur.* (2002) 148–160.
- [35] Shah T., Venkatesan S. Authentication of IoT Device and IoT Server Using Secure Vaults[C]// 2018:819–824.
- [36] F. Zhu, P. Li, H. Xu, et al., A Lightweight RFID mutual authentication protocol with PUF[J], *Sensors* 19 (13) (2019) 2957–2978.
- [37] Z. Min, X. Yao, L. Hong, et al., Physical unclonable function based authentication protocol for unit IoT and ubiquitous IoT[C]. *International Conference on Identification*, IEEE Computer Society, 2016, pp. 179–184.
- [38] M.N. Aman, K.C. Chua, B. Sikdar, Mutual authentication in IoT systems using physical unclonable functions, *IEEE Internet Things J.* 4 (5) (2017) 1327–1340.
- [39] M.N. Aman, K.C. Chua, B. Sikdar, Mutual authentication in IoT systems using physical unclonable functions, *IEEE Internet Things J.* 4 (5) (2017) 1327–1340.
- [40] S. Banerjee, V. Odelu, A.K. Das, et al., Physically secure lightweight anonymous user authentication protocol for internet of things using physically unclonable functions, *IEEE Access* 7 (2019) 85627–85644.
- [41] P. Gope, A.K. Das, N. Kumar, et al., Lightweight and physically secure anonymous mutual authentication protocol for real-time data access in industrial wireless sensor networks, *IEEE Trans. Ind. Inform.* 15 (9) (2019) 4957–4968.
- [42] U. Chatterjee, V. Govindan, R. Sadhukhan, et al., Building PUF based authentication and keyexchange protocol for iot without explicit crps in verifier database, *IEEE Trans. Dependable Secur. Comput.* 16 (3) (2019) 424–437.
- [43] P. Mall, R. Amin, A.K. Das, et al., PUF-based authentication and key agreement protocols for IoT, WSNs and smart grids: a comprehensive survey, *IEEE Internet Things J.* (2022) 1–1.
- [44] S. Challa, et al., Secure signature-based authenticated key establishment scheme for future IoT applications, *IEEE Access* 5 (2017) 3028–3043.
- [45] A.K. Sutrala, M.S. Obaidat, S. Saha, A.K. Das, M. Alazab, Y. Park, Authenticated key agreement scheme with user anonymity and untraceability for 5G-enabled softwarized industrial cyber-physical systems, *IEEE Trans. Intell. Transp. Syst.* 23 (3) (2022) 2316–2330.
- [46] M. Wazid, A.K. Das, N. Kumar, M. Alazab, Designing authenticated key management scheme in 6G-enabled network in a box deployed for industrial applications, *IEEE Trans. Ind. Inf.* 17 (10) (2021) 7174–7184.
- [47] M. Saqib, B. Jasra, A.H. Moon, A lightweight three factor authentication framework for IoT based critical applications, *J. King Saud. Univ. Comput. Inf. Sci.* 34 (9) (2022) 6925–6937.
- [48] H. Xu, C. Hsu, L. Harn, et al., Three-factor anonymous authentication and key agreement based on fuzzy biological extraction for Industrial Internet of Things, *IEEE Trans. Serv. Comput.* 16 (4) (2023) 3000–3013.